LEC3 - The Probability Set Function

Random Experiment: An act or process of observation that leads to a single outcome which cannot be predicted with certainty

Sample Space: The set of all possible outcomes, denoted by S

Simple Event: An event that cannot be decomposed. Each simple event corresponds to one and only one sample point

Sample Point: The most basic outcome of a random experiment

Compound Event: Consists of two or more simple events

Example

In the six-sided die tossing experiment:

- $S = \{e_1, e_2, e_3, e_4, e_5, e_6\}$, where e_i represents observing number i
- e denotes a simple event or corresponding sample point
- "observe a 1", "observe a 2", etc. are simple events because they cannot be decomposed further. Each simple event corresponds to 1 sample point
- "observe an odd number" is a compound event composed of e_1, e_3, e_5

Discrete Sample Space: A sample space that contains countable (finite or infinite) number of distinct sample points (ex. flip coin until you get heads is infinite but countable)

Event: A collection of sample points (subset of S)

Probability Axioms

1. The relative frequency of any event must be greater or equal to zero ($P(A) \geq 0$)
   - Negative frequency does not make sense
2. The relative frequency of the whole sample space must be 1 ($P(S) = 1$)
   - Something in S must occur every time an experiment is conducted
3. If two events are mutually exclusive, then the probability of their union is the sum of their respective frequencies

Theorem 7: The addition law of probability

The probability of the union of 2 events A and B is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

Proof:

$$A \cup B = \underbrace{A \cup (B \cap A^c)}^{\text{mutually exclusive}}$$

$$P(A \cup B) = P(A \cup (B \cap A^c))$$

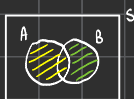
$$P(A \cup B) = P(A) + P(B \cap A^c), \text{ by axiom 3}$$

$$B = (A \cap B) \cup (B \cap A^c)$$

$$P(B) = P(A \cap B) + P(B \cap A^c) \Rightarrow P(B) - P(A \cap B) = P(B \cap A^c)$$

sub in

Therefore, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, as required



Theorem 8

For an event A, $P(A) = 1 - P(A^c)$

Example

Gene expression profiling is a state-of-the-art method for determining the biology of cells. In Briefings in Functional Genomics and Proteomics (Dec. 2006), biologists at Pacific Northwest National Library reviewed several gene expression profiling methods. The biologists applied two of the methods (A and B) to data collected on proteins in human mammary cells.

The probability that the protein is identified by method A is 0.41, the probability that the protein is identified by method B is 0.42, and the probability that the protein is identified by both is 0.40.

1. Find the probability that the protein is identified by either method A or B

   $A = \{y, n\}$, $P(y) = 0.41$ $P(n) = 0.59$ $B = \{y, n\}$, $P(y) = 0.42$, $P(n) = 0.58$ $P(A \cap B) = 0.4$

   $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

   $= 0.41 + 0.42 - 0.4 = 0.43$

2. Find the probability that the protein is not identified by either method

   $P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.43 = 0.57$

3. Find the probability that the protein is identified by exactly 1 method

   $P(A \oplus B) = P(A \cup B) - P(A \cap B) = 0.43 - 0.40 = 0.03$

4. Find the probability

   $P(A \cap B^c) = P(A) - P(A \cap B) = 0.41 - 0.40 = 0.01$